

KYPTONIX TECHNOLOGIES

Product Requirements Document

Corporate Website • IT Solutions & Software Development

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1. Executive Summary

This Product Requirements Document (PRD) defines the complete specification for the design, development, and launch of the official corporate website for Kyptonix Technologies. The website will serve as the company's primary digital presence — communicating its value proposition, showcasing its breadth of IT and software development services, generating qualified leads, and establishing trust with prospective clients across Africa and globally.

The website must be designed to serve a diverse audience ranging from startups and SMEs to enterprise clients and government institutions. No visitor should arrive at the site and fail to find a relevant solution. This document should be treated as the single source of truth and must be reviewed and approved by all project stakeholders before development commences.

2. Company Overview

Company Name	Kyptonix Technologies
Industry	Information Technology & Software Development
Target Markets	Kenya, East Africa, Pan-Africa, Global
Business Model	B2B, B2G (Government), and select B2C engagements
Core Value Prop	Reliable, innovative, and affordable IT solutions that accelerate business growth
Tone of Voice	Professional, forward-thinking, approachable, solution-oriented

2.1 Mission Statement

To empower businesses, governments, and communities through cutting-edge technology solutions that solve real-world problems and drive sustainable growth.

2.2 Vision Statement

To be East Africa's most trusted technology partner — known for excellence, reliability, and innovation.

3. Project Objectives

The website must achieve the following strategic objectives:

#	Objective	Success Metric
1	Establish credibility and brand authority online	Industry recognition, media mentions, backlinks
2	Present the widest possible range of IT services clearly	Visitor able to identify a relevant service within 30 seconds
3	Generate qualified inbound leads	Contact form submissions, quote requests per month
4	Showcase portfolio and client success stories	Portfolio views, case study downloads
5	Enable self-service information access	Reduced support inquiries, FAQ engagement
6	Support talent acquisition	Job applications received through the careers page
7	Rank on search engines for target keywords	Page 1 ranking for 10+ target keywords within 6 months
8	Facilitate partner and vendor engagement	Partner inquiry form submissions

4. Target Audience & User Personas

4.1 Primary Audiences

Persona	Description	Primary Goal on Site
SME Business Owner	Owner of a small/medium business looking for IT support, software, or website development	Find affordable, relevant services and make contact
Enterprise IT Manager	Technical decision-maker at a large company evaluating vendors	Assess technical capabilities, view portfolio, request proposal
Government/Procurement Officer	Public sector official sourcing IT solutions or systems	Verify credibility, review past projects, download company profile
Startup Founder	Early-stage founder needing MVP development, cloud, or tech advisory	Understand available services and get a quick quote
Non-Profit / NGO	Organisation seeking affordable tech solutions for social impact	Learn about relevant services and corporate social responsibility

Persona	Description	Primary Goal on Site
Potential Employee / Intern	Tech talent looking for a career opportunity	Browse open roles and understand company culture
Strategic Partner / Vendor	Another tech company exploring partnership or referral agreements	Understand services and initiate partner discussions
Investor / Funder	Individual or institution assessing Kyptonix for investment	Review company credentials, leadership, and financials

5. Website Architecture & Sitemap

The website shall be structured around a clear, SEO-optimised hierarchy. Every top-level page should be accessible within one click from the homepage. The full sitemap is defined below.

Level 1 (Page)	Level 2 (Sub-pages / Sections)	Priority
Home	Hero, Services Overview, About Snapshot, Featured Work, Testimonials, CTA, Blog Teaser, Partners	P0
About Us	Company Story, Mission & Vision, Core Values, Leadership Team, Awards & Recognition, Company Profile Download	P0
Services	Overview landing page — links to all sub-service pages	P0
→ Software Development	Custom Software, Mobile Apps, Web Applications, API Development, SaaS Products	P0
→ IT Infrastructure	Networking, Server Setup, Hardware Supply, Data Centres, IT Audits	P0
→ Cloud Services	Cloud Migration, Cloud Hosting, Managed Cloud, Backup & DR, Multi-Cloud	P0
→ Cybersecurity	Security Audits, Penetration Testing, Endpoint Security, Compliance, SOC Services	P0
→ IT Support & Managed Services	Helpdesk, Remote IT Support, IT Outsourcing, SLA Contracts	P0
→ Digital Transformation	Business Process Automation, ERP/CRM, AI & ML Solutions, Data Analytics, Digitalisation Advisory	P0
→ Web & Digital	Website Design, E-commerce, UI/UX Design, SEO, Social Media Management, Digital Marketing	P0
→ Enterprise Solutions	ERP Systems, CRM Platforms, HRMS, POS Systems, Inventory Management	P0
→ IoT & Emerging Tech	IoT Solutions, Smart Systems, Blockchain, AR/VR Solutions	P1
→ Training & Capacity Building	Corporate ICT Training, Certification Prep, Tech Workshops, School STEM Programs	P1
→ Government & Public Sector	E-Government Solutions, Civic Tech, Public Infrastructure, IFMIS/GHRIS integrations	P1
→ Startup & SME Package	Starter Packages, MVP Development, Tech Advisory, Affordable Bundles	P1
Portfolio	All Projects, Filter by Industry/Service, Featured Case Studies, Testimonials	P0
Case Studies	Individual detailed case study pages	P1

Level 1 (Page)	Level 2 (Sub-pages / Sections)	Priority
Blog / Resources	Articles, How-To Guides, Tech News, Whitepapers, E-books, Webinars	P1
Careers	Open Positions, Life at Kyptonix, Benefits, Application Form, Internship Program	P1
Partners	Technology Partners, Certifications, Partner Program, Vendor Registration	P1
Contact Us	Contact Form, Physical Address + Map, Phone/Email, Support Ticket, Live Chat	P0
Get a Quote	Service Selection, Project Brief Form, Budget Range, Timeline	P0
Client Portal (Login)	Project tracking, invoice access, support tickets (for existing clients)	P2
Legal Pages	Privacy Policy, Terms of Service, Cookie Policy, GDPR Notice	P0
404 Error Page	Friendly error page with navigation and search	P0

6. Page-by-Page Content Requirements

6.1 Homepage

The homepage is the single most important page. It must immediately communicate who Kyptonix is, what they do, and why visitors should trust them — in under 5 seconds.

6.1.1 Hero Section

- Full-width hero with background image or animated video loop (office/tech environment)
- Primary headline: powerful, benefit-driven (e.g., 'The Technology Partner That Powers Your Growth')
- Sub-headline: brief descriptor of what Kyptonix does
- Two CTAs: 'Explore Our Services' (primary) and 'Get a Free Quote' (secondary)
- Optional: animated counter stats — years in business, clients served, projects delivered, uptime SLA

6.1.2 Services Snapshot

- Grid or card layout showing 8–12 primary service categories with icons
- Each card links to the respective services sub-page
- Section headline: 'End-to-End Technology Solutions'

6.1.3 Why Choose Kyptonix

- 4–6 differentiators with icons: e.g., Expert Team, 24/7 Support, Scalable Solutions, Proven Track Record, Local Expertise, Global Standards

6.1.4 Featured Portfolio

- 3–4 highlighted projects with image, client name, service type, brief outcome
- 'View All Projects' CTA

6.1.5 Testimonials / Social Proof

- Rotating carousel of client testimonials with name, company, and photo
- Optional: logos of key clients in a horizontal scrolling strip

6.1.6 Call to Action Banner

- Full-width contrasting band: 'Ready to transform your business? Let's talk.'
- Single CTA button linking to Contact page

6.1.7 Blog / Insights Teaser

- 3 latest blog posts with thumbnail, title, date, and excerpt
- 'Read More Articles' link

6.1.8 Partner Logos Strip

- Logos of technology partners (Microsoft, Google, AWS, etc.) to build credibility

6.2 About Us Page

- Company story — founding, milestones, growth narrative
- Mission, Vision, and Core Values section with icons
- Leadership team profiles with photos, names, titles, and LinkedIn links
- Timeline infographic of company milestones
- Awards, certifications, and recognitions section
- Downloadable Company Profile (PDF)
- Embedded video: company culture or CEO welcome message (optional at launch)

6.3 Services Landing Page

- Visual grid of all service categories (12+) with icons and brief descriptions
- Each card is clickable and leads to a dedicated service sub-page
- Search/filter capability: filter services by category (Infrastructure, Software, Security, etc.)
- Sticky sidebar or top-filter bar for navigation
- CTA: 'Not sure what you need? Talk to our experts' with a chat or form link

6.4 Individual Service Pages (Template)

Each service sub-page (e.g., 'Custom Software Development') must follow this standard template:

- Hero section with service-specific headline, sub-text, and CTA
- What is this service? (Plain language overview)
- Key offerings within this service (bulleted, with icons)
- Our process / How we work (numbered steps or visual flow)
- Who is this for? (industries or business types)
- Technologies / Tools we use (logos or tags)
- Benefits & ROI (value proposition)
- Related case studies or portfolio samples

- FAQs specific to this service
- CTA section: 'Get a Quote' / 'Book a Consultation'

6.5 Portfolio / Projects Page

- Filterable project gallery (by service type, industry, region)
- Each project card: thumbnail, project name, client (if permitted), service category, brief outcome
- Clicking a card opens a detailed project/case study page
- Testimonial snippet from the client (where available)

6.6 Blog & Resources

- Category-based blog with pagination
- Featured post at the top
- Resources section: whitepapers, e-books, webinars, how-to guides (gated with email)
- Search functionality
- Social share buttons
- Author bio on each post
- Related posts at the end of each article

6.7 Careers Page

- Current job openings — listed with role title, department, location (remote/onsite), and date posted
- Filtering by department, location, or job type
- Each listing links to a detailed job description page with application form
- 'Life at Kyptonix' section with photos and culture highlights
- Benefits and perks summary
- Internship & Graduate Programme section
- General application form for unsolicited applications

6.8 Contact Page

- General contact form: Name, Email, Phone, Company, Subject, Message, File attachment
- Physical address with embedded Google Maps
- Phone and email contacts (separated by department: Sales, Support, HR, Finance)
- Office hours clearly stated

- WhatsApp quick contact button
- Support ticket submission portal (for existing clients)
- Live chat integration (e.g., Intercom, Tawk.to)

6.9 Get a Quote Page

- Multi-step wizard form:
 - Step 1 — Select services needed (multi-select from full catalogue)
 - Step 2 — Describe your project / brief
 - Step 3 — Timeline and budget range selection
 - Step 4 — Contact details + preferred contact method
- Confirmation email auto-sent after submission
- Sales team notified instantly via CRM/email

7. Services Catalogue

The following is the complete range of services that must be discoverable and described on the Kyptonix Technologies website. Each service category has its own dedicated page and may contain multiple sub-services.

7.1 Software Development

Sub-Service	Description
Custom Software Development	Bespoke software solutions built to unique business specifications — desktop, web, or cross-platform
Mobile Application Development	Native and cross-platform mobile apps for Android and iOS (React Native, Flutter, Kotlin, Swift)
Web Application Development	Responsive, scalable web apps using modern frameworks (React, Angular, Vue, Node.js, Django)
API Development & Integration	RESTful and GraphQL APIs; third-party system integrations (payment gateways, ERPs, CRMs)
SaaS Product Development	End-to-end SaaS platform design, development, and scaling for startups and enterprises
Software Testing & QA	Manual and automated testing, regression testing, performance testing, and QA as a service
Legacy System Modernisation	Migration of outdated systems to modern technology stacks without losing critical data or functionality

7.2 IT Infrastructure & Networking

Sub-Service	Description
Network Design & Installation	Structured cabling, LAN/WAN setup, wireless networks, routers, switches, and firewalls
Server Setup & Administration	Physical and virtual server procurement, setup, configuration, and management
Hardware Supply & Maintenance	Supply of laptops, desktops, printers, peripherals, and all IT hardware with maintenance contracts
Data Centre Solutions	Data centre design, rack installation, cooling systems, power management, and co-location services
IT Infrastructure Audit	Comprehensive assessment of existing IT infrastructure with recommendations for improvement
Virtualisation	VMware, Hyper-V, and KVM-based virtualisation for cost-efficient resource utilisation

Sub-Service	Description
Unified Communications	VoIP systems, video conferencing infrastructure, collaboration tools (Teams, Zoom integration)

7.3 Cloud Services

Sub-Service	Description
Cloud Migration	Seamless migration of on-premise systems to AWS, Microsoft Azure, or Google Cloud
Cloud Hosting & Management	Scalable, managed hosting on leading cloud platforms with 99.9% uptime SLA
Managed Cloud Services	Ongoing cloud infrastructure management including monitoring, patching, and optimisation
Backup & Disaster Recovery	Automated cloud-based backup solutions and disaster recovery planning and testing
Multi-Cloud Strategy	Design and management of multi-cloud environments for resilience and vendor independence
Cloud Security	Identity and access management, cloud firewall, encryption, and compliance monitoring
Cloud Cost Optimisation	Analysis and restructuring of cloud usage to reduce cost without impacting performance

7.4 Cybersecurity

Sub-Service	Description
Security Audits & Assessments	Thorough evaluation of security posture across network, applications, and human processes
Penetration Testing	Ethical hacking exercises to identify and exploit vulnerabilities before attackers do
Endpoint Security	Antivirus, EDR, MDM, and device management solutions for laptops, mobiles, and IoT devices
Security Operations Centre (SOC)	24/7 threat monitoring, detection, and incident response services
Compliance & Governance	Support for ISO 27001, GDPR, Kenya Data Protection Act, PCI-DSS, and other regulatory frameworks
Security Awareness Training	Staff training programs on phishing, social engineering, password security, and best practices
Incident Response	Rapid response to security breaches including forensics, containment, and recovery

Sub-Service	Description
Firewall & VPN Management	Configuration, monitoring, and management of network perimeter security devices

7.5 IT Support & Managed Services

Sub-Service	Description
Helpdesk Support (L1/L2/L3)	Multi-tiered technical support via phone, email, chat, and remote desktop tools
Remote IT Support	Real-time remote troubleshooting and issue resolution for businesses anywhere
On-Site IT Support	Physical IT technician visits for hardware issues, setup, and urgent on-site tasks
IT Outsourcing	Full or partial outsourcing of IT department functions for cost savings and expertise access
SLA-Based Contracts	Service level agreements with guaranteed response times and performance benchmarks
IT Asset Management	Tracking, maintenance scheduling, procurement, and lifecycle management of IT assets
Proactive Monitoring	24/7 system, network, and application monitoring with automated alerts and remediation

7.6 Digital Transformation

Sub-Service	Description
Business Process Automation (BPA)	Automation of repetitive business workflows using RPA tools (UiPath, Power Automate)
Enterprise Resource Planning (ERP)	Implementation and customisation of ERP systems (Odoo, SAP, Oracle, Microsoft Dynamics)
Customer Relationship Management (CRM)	Salesforce, HubSpot, Zoho CRM — implementation, customisation, and training
AI & Machine Learning Solutions	Custom AI models, chatbots, recommendation engines, predictive analytics, and NLP solutions
Data Analytics & Business Intelligence	Power BI, Tableau, and custom dashboards for data-driven decision making
Digital Transformation Advisory	Strategic consulting to guide organisations through digital change management
Document Management Systems	Paperless office solutions with workflow automation, version control, and e-signatures

7.7 Web & Digital Solutions

Sub-Service	Description
Website Design & Development	Professional, responsive, SEO-friendly websites for businesses of all sizes
E-Commerce Solutions	Fully functional online stores with payment gateway integration (M-Pesa, Visa, PayPal)
UI/UX Design	User-centred design for websites, mobile apps, and software — wireframes to high-fidelity prototypes
Search Engine Optimisation (SEO)	On-page, off-page, and technical SEO to improve organic search visibility
Digital Marketing	Google Ads, Facebook/Instagram Ads, email marketing, and content marketing campaigns
Social Media Management	Content creation, scheduling, community management, and performance reporting
Brand Identity & Graphics	Logo design, brand guidelines, marketing materials, and corporate identity packages

7.8 Enterprise Solutions

Sub-Service	Description
Enterprise Resource Planning (ERP)	Full-scale ERP deployment for large organisations — finance, HR, procurement, and operations
Human Resource Management System (HRMS)	Payroll, leave management, performance appraisal, and employee self-service portals
Point of Sale (POS) Systems	Hardware and software POS solutions for retail, hospitality, and service industries
Inventory & Warehouse Management	Real-time stock tracking, reorder automation, and warehouse management systems
Fleet Management Systems	GPS tracking, fuel management, maintenance scheduling, and driver behaviour monitoring
School Management Systems	Student information, fees management, timetabling, parent portal, and academic reporting
Hospital/Health Management Systems	Electronic health records (EHR), appointment booking, billing, and pharmacy management
Property Management Systems	Tenant management, rent collection, maintenance tracking for real estate companies

7.9 IoT & Emerging Technologies

Sub-Service	Description
IoT Solution Development	Design and deployment of connected device solutions for smart homes, agriculture, and industry
Smart Building Systems	Automated lighting, HVAC, access control, and energy management systems
Blockchain Development	Custom blockchain solutions for supply chain, finance, identity, and document verification
Augmented & Virtual Reality (AR/VR)	Immersive experiences for training, real estate, retail, and education
Drone Solutions	Drone procurement, configuration, and integration for agriculture, surveying, and logistics

7.10 Training & Capacity Building

Sub-Service	Description
Corporate ICT Training	Custom training programs for organisations on productivity software, cybersecurity, and IT tools
Professional Certification Preparation	Training for CompTIA, Cisco (CCNA), Microsoft (Azure, M365), AWS, and Google certifications
Software User Training	End-user training on specific software solutions deployed by Kyptonix
Tech Bootcamps & Workshops	Short, intensive programs for developers, designers, and tech entrepreneurs
School & University STEM Programs	ICT lab setup, STEM curriculum support, and student coding programs

7.11 Government & Public Sector

Sub-Service	Description
E-Government Solutions	Citizen-facing digital platforms for public service delivery (permits, licensing, payments)
Civic Technology	Open data portals, public consultation platforms, and government transparency tools
Public Infrastructure IT	ICT infrastructure projects for county governments, schools, hospitals, and public spaces
Government Systems Integration	Integration with national systems — KRA iTax, NHIF, NSSF, eCitizen, IFMIS, GHRIS

Sub-Service	Description
GIS & Mapping Solutions	Geographic information systems for land management, urban planning, and resource tracking

7.12 Startup & SME Packages

Sub-Service	Description
Starter Tech Bundle	Affordable package: website + email hosting + basic IT support for new businesses
MVP Development	Rapid prototyping and Minimum Viable Product development for startups
Tech Advisory Services	Part-time CTO / tech advisor service for startups without in-house technical leadership
Business Automation Starter	Entry-level process automation tools for SMEs to reduce manual workload
Growth Marketing Package	SEO + social media + Google Ads combo package for growing SMEs

8. Functional Requirements

8.1 Core Features (Must Have — P0)

Feature	Description	Priority
Responsive Design	Fully responsive across mobile, tablet, and desktop (min. 320px to 2560px)	P0
Navigation	Sticky header with mega-menu for services, hamburger menu on mobile	P0
Contact Form	Multi-field contact form with validation, CAPTCHA, and email notification	P0
Quote Request Wizard	Multi-step project brief form with service selection, budget, and timeline fields	P0
Blog / CMS	Content Management System for blog posts with categories, tags, and author profiles	P0
Portfolio Gallery	Filterable project gallery with detailed project pages	P0
SEO Foundation	Meta titles, descriptions, Open Graph tags, structured data, sitemap.xml, robots.txt	P0
Cookie Consent	GDPR-compliant cookie consent banner with accept/reject options	P0
Live Chat Widget	Embedded live chat (Tawk.to or Intercom) with offline messaging fallback	P0
WhatsApp Button	Floating WhatsApp contact button on all pages	P0
Social Media Links	Header/footer links to LinkedIn, Twitter/X, Facebook, Instagram, YouTube	P0
Company Profile Download	PDF download of company profile available from About and Contact pages	P0
404 & Error Pages	Custom, branded 404 and 500 error pages with helpful navigation links	P0
Google Analytics	GA4 integration with event tracking for forms, downloads, and CTA clicks	P0
Back to Top Button	Sticky 'back to top' button visible on scroll on all long pages	P0

8.2 Secondary Features (Should Have — P1)

- Client Portal — login-protected area for project updates, invoices, and support tickets
- Case Study pages — detailed storytelling pages with problem, solution, and results sections

- Careers module — job listings with filters and integrated application forms
- Newsletter subscription — opt-in form with MailChimp / Sendinblue integration
- Resource library — gated content (whitepapers, e-books) behind email capture forms
- Testimonials management — admin ability to add/remove/order testimonials
- Project ROI calculator — interactive tool for estimating ROI on tech investments

8.3 Future Features (Nice to Have — P2)

- AI-powered chat assistant trained on Kyptonix services
- Client self-service onboarding portal
- Partner portal with referral tracking
- Multi-language support (Swahili, French)
- Mobile app version of client portal

9. Non-Functional Requirements

Requirement	Specification
Performance	Page load time $\leq$ 3 seconds on standard broadband; $\leq$ 5 seconds on 3G mobile
Uptime / Availability	99.9% uptime SLA. Scheduled maintenance communicated in advance
Scalability	Architecture must support 10x traffic growth without redesign
Security	HTTPS/SSL enforced site-wide. Regular security scanning. WAF protection
Browser Compatibility	Chrome, Firefox, Safari, Edge — last 2 major versions each
Mobile-First	Design and development approach must prioritise mobile experience
Accessibility	WCAG 2.1 Level AA compliance minimum
SEO Technical	Core Web Vitals green scores. Crawlable site structure. Canonical URLs
Data Privacy	Kenya Data Protection Act 2019 and GDPR-aligned data handling
Backup	Daily automated backups retained for minimum 30 days
CMS Usability	Non-technical staff must be able to update content without developer involvement
Internationalisation	Codebase ready for multi-language support in Phase 2

10. Design & Brand Guidelines

10.1 Visual Identity

Element	Specification
Aesthetic	Modern, clean, professional. Dark navy tones balanced with bright accent colours
Primary Colour	#1A3A5C (Deep Navy Blue) — conveys trust, expertise, and stability
Secondary Colour	#4A90D9 (Tech Blue) — for accents, buttons, and highlights
Accent Colour	#00C6AE (Teal/Green) — for CTAs, success states, and energy
Background	#F2F6FA (Light Grey-Blue) for sections; #FFFFFF for content areas
Error / Alert	#E74C3C (Red) for errors; #27AE60 (Green) for success messages
Primary Font	Inter or DM Sans — clean, geometric, highly readable at all sizes

Element	Specification
Heading Font	Poppins or Montserrat — bold, modern, authoritative for headings
Monospace Font	JetBrains Mono or Fira Code — for code snippets in blog/resources
Logo Usage	Logo provided by Kyptonix. Dark and light variants required. Minimum clear space: 1x logo height on all sides
Imagery Style	High-quality tech photography, abstract digital visuals, and diverse African professionals. No stock-cliché photos
Iconography	Outlined icons (Phosphor Icons or Lucide). Consistent 24px grid. No filled icons mixed with outlined
Illustration Style	Flat vector illustrations used sparingly to explain complex concepts
Animation	Subtle scroll-triggered animations (fade-in, slide-up). No autoplay video with sound. No excessive motion

10.2 UX Principles

- Clarity over cleverness — every element must serve a functional purpose
- Progressive disclosure — show high-level first, allow users to drill down
- Consistent navigation — users should always know where they are and how to proceed
- Trust signals on every page — certifications, testimonials, guarantees
- Every page has a CTA — no dead ends; always give the user a next step

11. Technology Stack

The following is the recommended technology stack. Final selection may be adjusted based on developer preferences and budget, but must maintain the non-functional requirements defined in Section 9.

Layer	Recommended Technology	Alternatives
Frontend Framework	Next.js (React)	Gatsby, Astro
CMS	Sanity.io or Contentful (Headless CMS)	WordPress (Headless), Strapi
Styling	Tailwind CSS	CSS Modules, Styled Components
Hosting	Vercel, Netlify, or AWS Amplify	DigitalOcean, Linode, Azure Static Web Apps
Database	PostgreSQL (for portal/forms)	MySQL, MongoDB
Backend / API	Node.js + Express or Next.js API Routes	Django REST Framework, Laravel
Email Service	SendGrid or Postmark (transactional emails)	Mailgun, AWS SES
CRM Integration	HubSpot or Zoho CRM	Salesforce, Pipedrive
Analytics	Google Analytics 4 + Google Tag Manager	Mixpanel, Plausible
Live Chat	Tawk.to or Intercom	Crisp, Zendesk Chat
Forms	React Hook Form + Zod validation	Formik, custom forms
Image CDN	Cloudinary or imgix	Next.js Image, AWS CloudFront
SSL/Security	Let's Encrypt (auto-renewing)	Cloudflare SSL
Version Control	GitHub (private repository)	GitLab, Bitbucket
CI/CD	GitHub Actions	CircleCI, Vercel auto-deploy

12. SEO & Content Strategy

12.1 Technical SEO Requirements

- All pages must have unique, optimised meta titles (50–60 chars) and meta descriptions (150–160 chars)
- Canonical URLs on all pages to prevent duplicate content
- XML sitemap auto-generated and submitted to Google Search Console

- Robots.txt configured to allow crawling of public pages, block admin/portal
- Structured data markup (JSON-LD) for: Organisation, LocalBusiness, Service, FAQPage, Article
- Image alt text required on all images
- Core Web Vitals optimisation: LCP < 2.5s, FID < 100ms, CLS < 0.1
- Breadcrumb navigation on all interior pages

12.2 Content Strategy

- Blog publishing cadence: minimum 2 posts per month at launch; 4 per month from Month 3
- Blog topics must include: IT industry news, how-to guides, case study summaries, product comparisons, Kenya/Africa tech trends
- Each service page must target 1 primary keyword and 2–3 secondary keywords
- All content must pass a readability score of Flesch-Kincaid Grade ≤ 10
- Internal linking strategy: each blog post must link to at least 2 service pages

13. Accessibility & Compliance

Requirement	Standard / Detail
Colour Contrast	Minimum 4.5:1 contrast ratio for body text; 3:1 for large text and UI components
Keyboard Navigation	All interactive elements must be operable via keyboard (Tab, Enter, Escape)
Screen Reader Support	Semantic HTML5, ARIA labels on icons and form fields, skip-to-content link
Font Size	Minimum 16px base font size. No text below 12px on any element
Focus Indicators	Visible focus rings on all focusable elements
Form Labels	All form inputs must have associated visible labels
Error Messages	Form validation errors must be descriptive and associated with specific fields
Media Captions	All videos must have captions/subtitles
Kenya Data Protection Act	Privacy policy must detail data collection, use, and user rights under KDPA 2019
GDPR	Cookie consent management; right to erasure; data processing agreement available on request
WCAG Level	Minimum WCAG 2.1 Level AA. Aim for AAA where feasible

14. Integrations & Third-Party Tools

Integration	Purpose	Priority
Google Analytics 4	Website traffic analysis, user behaviour, and conversion tracking	P0
Google Tag Manager	Centralised tag management without code deployments	P0
Google Search Console	SEO monitoring, crawl error detection, and search performance	P0
HubSpot / Zoho CRM	Lead capture from forms, lead scoring, and sales pipeline management	P0
Tawk.to / Intercom	Live chat with visitor messaging and lead capture	P0
SendGrid / Postmark	Transactional email delivery (form confirmations, alerts)	P0

Integration	Purpose	Priority
WhatsApp Business API	Floating WhatsApp contact button and automated welcome message	P0
Cloudinary / imgix	Image CDN for optimised, responsive image delivery	P0
MailChimp / Brevo	Email newsletter campaigns and subscriber management	P1
Calendly / Cal.com	Embedded meeting booking for consultation scheduling	P1
Stripe / M-Pesa Daraja API	Online payment for services or deposit collection (Phase 2)	P2
Slack	Internal notifications for new leads and form submissions	P1
Hotjar / Microsoft Clarity	Heatmaps, session recordings, and UX behaviour analysis	P1

15. Analytics & Key Performance Indicators

KPI	Target (Month 6)	Measurement Tool
Monthly Website Sessions	5,000+	Google Analytics
Bounce Rate	< 55%	Google Analytics
Average Session Duration	> 2 minutes	Google Analytics
Contact Form Submissions	50+ per month	GA4 Events + CRM
Quote Requests Submitted	20+ per month	GA4 Events + CRM
Blog Newsletter Subscribers	500+	MailChimp
Organic Search Traffic Share	> 40% of total sessions	Google Analytics
Google Search Console Clicks	1,000+ per month	Search Console
Core Web Vitals	All green	PageSpeed Insights
Job Applications Received	10+ per month	Careers Form Analytics
Company Profile Downloads	30+ per month	GA4 File Download Events
Live Chat Conversations	25+ per month	Tawk.to / Intercom Dashboard

16. Content Governance

16.1 Content Ownership

Content Area	Owner / Responsible Party
Homepage & Core Pages	Marketing Lead + CEO final approval
Service Pages	Product/Services Team + Technical Lead review
Blog & Resources	Content Manager (or outsourced content team)
Portfolio & Case Studies	Project Managers + Client Approval required before publishing
Careers	HR Department
Legal Pages	Legal Counsel / Company Secretary
CMS Maintenance	Website Administrator / Designated Developer

16.2 Update Frequency

- Services pages: reviewed and updated quarterly
- Blog: minimum 2 posts per month
- Portfolio: new project added within 2 weeks of project completion and client approval
- Team page: updated within 1 week of staff changes
- Careers: job postings updated in real time as positions open/close
- Legal pages: reviewed annually or upon regulatory changes

17. Security Requirements

Requirement	Specification
SSL/HTTPS	SSL enforced on all pages and subdomains. HTTP auto-redirects to HTTPS
Web Application Firewall (WAF)	Cloudflare WAF or equivalent — DDoS mitigation, bot protection, rate limiting
Form Security	CAPTCHA (Google reCAPTCHA v3) on all public-facing forms to prevent spam
Dependency Management	All third-party libraries kept updated. Monthly vulnerability scanning (Dependabot or Snyk)
Admin Access Control	CMS admin protected by strong password policy and Two-Factor Authentication (2FA)
Data Transmission	All form data transmitted over TLS 1.2+. No sensitive data stored in URL parameters
File Upload Security	File type validation, size limits, and malware scanning on any file upload features
Content Security Policy (CSP)	HTTP security headers: CSP, X-Content-Type-Options, X-Frame-Options, HSTS
Penetration Testing	Website penetration test conducted before launch and annually thereafter
Backup & Recovery	Daily automated backups. RTO (Recovery Time Objective): < 4 hours. RPO: < 24 hours

18. Milestones & Timeline

The following timeline assumes a dedicated team beginning work immediately upon PRD approval. Timelines are indicative and subject to revision based on team capacity and stakeholder review cycles.

Phase	Activities	Duration	Deliverable
Phase 1: Discovery & Planning	Stakeholder interviews, content audit, competitor analysis, keyword research, final sitemap confirmation	1–2 weeks	Approved sitemap, keyword list, content brief
Phase 2: Design	Wireframes (all pages), UI design (all pages), brand application, prototype, stakeholder review & revisions	2–3 weeks	Approved Figma prototype

Phase	Activities	Duration	Deliverable
Phase 3: Development	Frontend development, CMS setup, backend/API, integrations (CRM, chat, analytics), form logic, performance optimisation	4–6 weeks	Staging site ready for testing
Phase 4: Content Population	Copy writing and review, image sourcing/optimisation, blog posts (first 4), case studies (first 3)	2–3 weeks	All pages fully populated on staging
Phase 5: Testing & QA	Cross-browser testing, mobile testing, accessibility audit, SEO audit, form testing, load testing, security scan	1–2 weeks	QA report — all critical bugs resolved
Phase 6: Launch	DNS cutover, SSL verification, analytics verification, Search Console setup, sitemap submission, post-launch monitoring	1 week	Live website
Phase 7: Post-Launch Optimisation	Analytics review, CRO experiments, content additions, bug fixes, SEO improvements	Ongoing	Monthly performance report

19. Risks & Mitigations

Risk	Likelihood	Impact	Mitigation
Content delays (copy not ready on time)	High	High	Assign dedicated content owner; use placeholder content in dev; set hard content deadlines
Scope creep during development	High	Medium	Freeze scope after PRD sign-off; all new requests go to a change request log for Phase 2
Third-party integration failures (CRM, chat, email)	Medium	Medium	Test all integrations in staging; maintain fallback (direct email) for lead capture
Slow page load on African mobile networks	Medium	High	Aggressive image optimisation, CDN usage, lazy loading, AMP consideration for blog
Security vulnerabilities post-launch	Low	Very High	WAF, pre-launch pen test, dependency monitoring, and rapid patch process in place
Low organic traffic in first 3 months	High	Medium	Paid search ads to supplement organic while SEO matures; aggressive content publishing
Key developer unavailability mid-project	Low	High	Code documentation, version control, and at least 2 developers familiar with the codebase

20. Glossary

Term	Definition
CMS	Content Management System — software for creating and managing website content without coding
CRO	Conversion Rate Optimisation — the practice of improving website performance to increase % of visitors who complete a goal
CTA	Call to Action — a button, link, or prompt that encourages a specific user action
DR	Disaster Recovery — processes for restoring IT systems after a failure or security incident
ERP	Enterprise Resource Planning — integrated software managing core business processes
GDPR	General Data Protection Regulation — EU law governing data privacy and personal information handling
HRMS	Human Resource Management System — software for managing employee data, payroll, and HR processes
IoT	Internet of Things — network of physical devices embedded with sensors and internet connectivity
KDPA	Kenya Data Protection Act 2019 — Kenyan legislation governing personal data collection and processing
KPI	Key Performance Indicator — measurable metric used to evaluate success against an objective
MVP	Minimum Viable Product — a product with enough features to attract early adopters and validate a business idea
P0/P1/P2	Priority levels — P0: Must have, P1: Should have, P2: Nice to have
POS	Point of Sale — system for processing sales transactions at the point of purchase
PRD	Product Requirements Document — this document; defines what a product must do and how it should behave
RPA	Robotic Process Automation — technology to automate repetitive rule-based digital tasks
SEO	Search Engine Optimisation — practice of improving website visibility in search engine results
SLA	Service Level Agreement — contract defining the expected level of service between provider and client
SOC	Security Operations Centre — centralised unit for monitoring and responding to cybersecurity threats
UI/UX	User Interface / User Experience — the design and feel of digital products and how users interact with them

Term	Definition
WAF	Web Application Firewall — security system filtering and monitoring HTTP traffic to/from a web application
WCAG	Web Content Accessibility Guidelines — international standards for making web content accessible to all users

21. Approval & Sign-Off

By signing below, each stakeholder acknowledges that they have reviewed this Product Requirements Document and approve it as the authoritative specification for the Kyptonix Technologies website project. Any changes to requirements after this sign-off must follow the formal change request process.

Role	Name	Signature	Date
CEO / Founder		_____	_____
Technical Lead / CTO		_____	_____
Marketing Lead		_____	_____
Project Manager		_____	_____
Head of Design		_____	_____
Legal / Compliance		_____	_____
External Reviewer (optional)		_____	_____

— End of Document —

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